Study of the radioactive decays of ^{140}Ba and ^{140}La

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The intensities of X rays and γ rays from the decays of ^{140}Ba and ^{140}La were measured precisely using Si(Li) and HPGe detectors. The *L* X-ray intensities in ^{140}Ba decay are reported for the first time. The conversion electrons from these decays are investigated using a mini-orange electron spectrometer. The electron intensities for the (*M* + *N*.) conversion of 329, 487, 1596, and 1903 keV transitions in ^{140}Ce were measured for the first time. From the present conversion-electron and γ-ray intensities, the conversion coefficients for various transitions in ^{140}La and ^{140}Ce were determined. Also, the γ–γ directional correlations for 15 cascades in ^{140}Ce were studied using a HPGe–HPGe detector coincidence setup (time resolution = 7 ns). The 109–(329)– 487, 131–242, and 131–266 keV cascades in ^{140}Ce were studied for the first time. The multipole mixing ratios for the 109, 131, 242, 266, 329, 432, 487, 751, 816, 868, 919, 925, and 951 keV transitions in ^{140}Ce are deduced from the present directional correlation and conversion-coefficient measurements.

On a mesuré avec précision, en utilisant des détecteurs Si(Li) et HPGe, les intensités des rayons X et des rayons γ provenant des désintégrations de ^{140}Ba et ^{140}La. Les intensités des rayons X *L* dans la désintégration de ^{140}Ba sont rapportées pour la première fois. Les électrons de conversion provenant de ces désintégrations ont été étudiés en utilisant un spectromètre à électrons mini-orange. Les intensités électroniques pour la conversion (*M* + *N*) des transitions de 329, 487, 1596 et 1903 keV dans ^{140}Ce ont été mesurées pour la première fois. A partir de ces mesures des intensités d'électrons de conversion et de rayons γ, on a déterminé les coefficients de conversion pour diverses transitions dans ^{140}La et ^{140}Ce. On a aussi étudié les corrélations directionnelles γ–γ pour 15 cascades dans ^{140}Ce, en utilisant un arrangement de détecteurs HPGe–HPGe en coïncidence (résolution temporelle = 7 ns). Les cascades 109–(329)–487, 131–242 et 131–266 keV dans ^{140}Ce ont été étudiées pour la première fois. Les rapports de mélange multipolaire pour les transitions de 109, 131, 242, 266, 329, 432, 487, 751, 816, 866, 919, 925 et 951 keV dans ^{140}Ce ont été déduits des mesures de corrélation directionnelle et de coefficients de conversion.
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1. Introduction

The radionuclides ^{140}Ba ($T_{1/2}$ = 12.7 d) and ^{140}La ($T_{1/2}$ = 1.68 d) undergo β^- decay to the excited states of ^{140}La and ^{140}Ce, respectively. These excited states de-excite through the competing γ emission and internal conversion process. The study of the precise energy and intensity of the emitted γ rays and conversion electrons provides information on the nuclear levels. For a better understanding of subsequent atomic processes, there is a need for the measurement of the intensities of the X rays emitted following these decays. The level structure of ^{140}La and ^{140}Ce from the above mentioned decays has been extensively investigated by many research workers (1–18). These studies include the measurements of energies and intensities of γ rays (1–10), conversion electrons (5, 10), and multipole-mixing ratios (9–14) for the various transitions involved in these decays. Some conclusions about the excited states of ^{140}La and ^{140}Ce have also been drawn from reaction work (15, 16) and model calculations (17, 18). In spite of these investigations, uncertainties regarding the following features in these decays still prevail.

In ^{140}Ba decay, the earlier-measured intensities (2–4) for weak and low-energy γ rays are either available sparsely or found to be inconsistent. Recently, Adam *et al.* (10) measured the intensities of *K* X rays, γ rays, and conversion electrons from this decay and reported a new γ transition of 551.2 keV, which de-excites the 581 keV level and still needs to be confirmed. Kalinnikov *et al.* (4) measured the *K* X-ray and γ-ray intensities from ^{140}Ba decay using a poor energy-resolution Ge(Li) detector (resolution = 3.2 keV at 537.3 keV). The *K* X-ray intensities measured by them are found to be significantly higher than the evaluated results (19). To the best of our knowledge, no report of measured *L* X-ray intensities from the ^{140}Ba decay is available in the literature.

In the ^{140}La decay, γ-ray intensities have been reported by many workers (5–10). The available intensities for a few weak γ rays have been found to be inconsistent. Kaur *et al* (9) observed a new γ ray at 1303.5 keV and also proposed a new level at 3000.7 keV in ^{140}Ce. Adam *et al.* (10) reported new γ rays of energy 992.9 and 1877.3 keV and proposed three levels at 3394.9, 3473.6, and 3520.8 keV to fit the newly observed γ rays. The existence and placement of these γ rays and energy levels still needs to be confirmed. Also, the *K* X-ray intensities measured by Kalinnikov *et al.* (7) are significantly higher than the evaluated results (19).

Karlsson *et al.* (5) measured the intensities of conversion electrons from the ^{140}La decay using a double-focusing electron spectrometer, and the directional correlation measurements for different cascades in ^{140}Ce have been reported by many workers (9–14) using NaI (Tl) and Ge(Li) detector coincidence setups. The multipole mixing ratios of some of the transitions and the spin of the 2516 keV level in ^{140}Ce, determined on the basis of directional correlation and conversion-coefficient measurements by earlier workers, are found to be controversial.

In view of the above, it was thought worthwhile to reinvestigate the level structure of ^{140}La and ^{140}Ce precisely. The present work includes the intensity measurements of X rays and γ rays using well-calibrated high-resolution Si(Li) and HPGe detectors. We had a particular interest in investigating weak and low-energy γ rays. The conversion-electron spectra were studied using a mini-orange electron spectrometer. Also, γ–γ directional correlation measurements for various cascades in ^{140}Ce were carried out using a HPGe–HPGe detectors coincidence setup that had better energy and time resolution than those

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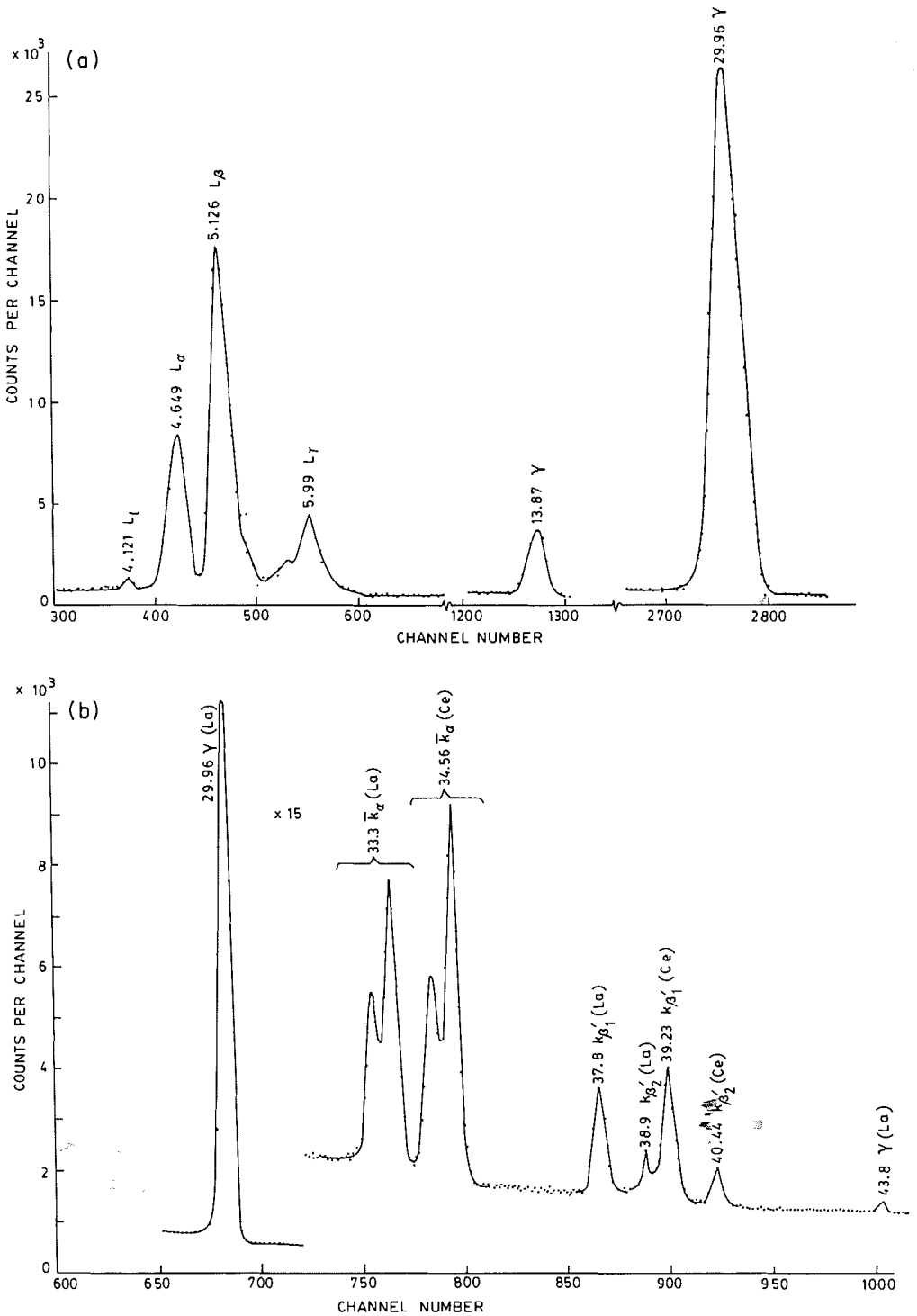


FIG. 1. Typical spectrum of ^{140}Ba decay: (*a*) The *L* X-ray part taken with the Si(Li) detector and (*b*) The *K* X-ray part taken with vertical HPGe detector.

used by previous workers (9–14). From these measurements, it has been possible to determine the multipole-mixing ratios of 13 transitions in ^{140}Ce.

2. Experimental methods and results

2.1. X-ray and γ-ray intensity measurements

The radioactive liquid sources of ^{140}Ba and ^{140}La in dilute HCl were obtained from Bhabha Atomic Research Centre, Bombay, India. Thin uncovered sources were prepared by drying the respective source solutions on Mylar backing for the measurements of X rays and low-energy γ rays. The count rates for these sources were kept at about 250 counts s^{-1} with the vertical HPGe and Si(Li) detectors. For intensity measurements of γ rays above 80 keV, the strength of the sources were so adjusted that they provided about 1200 counts s^{-1} when placed at a distance of 25 cm from the coaxial HPGe detectors.

The present X-ray and γ-ray intensity measurements were

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performed using a set of four semiconductor detectors; a horizontal planar Si(Li) detector (active volume = 28.27 mm^2 × 5.5 mm; FWHM = 165 eV at 5.9 keV), a vertical planar HPGe detector (active volume = 28.27 mm^2 × 5.0 mm; FWHM = 459 eV at 122 keV) and two coaxial HPGe detectors (active volumes = 57.1 cm^3 and 96.0 cm^3; FWHM = 1.7 keV at 1332 keV). These detectors were coupled to an 8K-channel analyser (ND 66B) through a spectroscopy amplifier (ORTEC 572).

The efficiency calibration curves for the coaxial HPGe detectors in the energy region 80–4000 keV, for the vertical planar HPGe detector in the energy region 20–600 keV, and for the Si(Li) detector in the energy region 4–90 keV were generated using standard radioactive sources as described in our earlier papers (20, 21).

Ten spectra were taken with each source and detector combination, for the time duration 25 000–150 000 s, which depended upon the counting statistics. The spectra in the X-ray regions were analyzed by adding the counts under the peaks, and background counts were subtracted by extrapolating the background counts above and below the peak region. The X-ray spectrum from the successive decay of $^{140}Ba \rightarrow {}^{140}La \rightarrow {}^{140}Ce$ obtained with Si(Li) and vertical HPGe detectors is shown in Fig. 1. It is clear from Fig. 1*b* that the *K* X rays from two decays are well separated except that the K_{β_2} peak of ^{140}La overlaps the K_{β_1} peak of ^{140}Ce. The area of K_{β_2} peak of ^{140}La was obtained by subtracting the area of the K_{β_1} peak of ^{140}Ce (measured from the ^{140}La decay separately) from the total area of the [K_{β_2} (La) + K_{β_1} (Ce)] peak. The contributions to the different *L* X rays from the ^{140}Ba decay due to *L* X rays from the ^{140}La decay were subtracted by using the intensity results from ref. 19 and were found to be less than 2%.

The γ-ray spectra were analyzed to obtain the area of different γ-ray peaks using the computer program SAMPO (22).

TABLE 1. Summing corrections for certain weak cross-over γ rays from ^{140}Ba and ^{140}La decays with a 96.0 cm^3 HPGe detector

γ ray (keV)	Summing γ rays ($\gamma_1 + \gamma_2$)	Estimated summing area ($\gamma_1 + \gamma_2$)	Observed area γ_3	%Summing correction
		^{140}Ba		
467.6	304.9 + 162.7	0.041	0.081(12)	−50.6%
		^{140}La		
1924.6	328.7 + 1596.2	0.018	0.031(2)	−58.1%
2083.2	487.0 + 1596.2	0.034	0.070(4)	−48.6%
2464.9	868.7 + 1596.2	0.0019	0.0089(9)	−21.3%

The areas of various cascading and cross-over γ rays were corrected for summing, using the formulation suggested by Gehrke *et al.* (8). The total efficiency curves for the coaxial HPGe detectors, required for an estimation of summing corrections, were obtained from the knowledge of the absorption coefficients for γ rays in germanium and the size of the detectors (23). The summing corrections in the case of measurements done using a 96.0 cm^3 coaxial HPGe detector, for example, are found to be 0.14 and 0.22% for the 162.7 and 304.9 keV γ rays in the ^{140}Ba decay and the 0.2, −1.7, 0.3, 0.3, 0.2, −0.3, and −0.3% for the 328.8, 397.5, 432.5, 487.0, 925.2, 2347.9, and 2547.3 keV γ rays in the ^{140}La decay. The estimated areas due to the summing of cascading γ rays for certain weak cross-over transitions, where summing corrections are found to be significantly large, are compared with the observed areas in Table 1.

The present measured X-ray and γ-ray intensities from ^{140}Ba and ^{140}La decays are shown in Tables 2 and 3. The intensity values for different γ rays as well as *K* X rays, measured with

TABLE 2. Relative intensities of X rays and γ rays in a ^{140}Ba decay

Energy (keV)	Relative intensities				
	This work	Ref. 8	Ref. 4	Ref. 10	Ref. 19
4.121 L_1	0.40(4)	—	—	—	0.49(9)
4.649 L_α	13.9(8)	—	—	—	13.8(25)
5.126 L_β	33.2(20)	—	—	—	31.6(57)
5.99 L_γ	6.6(5)	—	—	—	6.1(6)
13.87 γ	4.69(12)	—	7.2(25)	4.9(6)	4.9(6)
29.96 γ	58.4(10)	—	72(12)	60(3)	55.8(5)
33.3 $\overline{K}_\alpha$	6.10(18)	—	10.0(20)	6.5(5)	7.01(3)
37.8 K_{β_1}	1.15(5)	—	<2.3	1.60(5)	1.27(7)
38.9 K_{β_2}	0.32(5)	—			0.33(2)
43.8	0.054(7)	—	—	<0.007	0.062(4)
113.6	0.072(6)	—	—	0.077(16)	0.074
118.9	0.25(1)	—	0.21(2)	0.27(3)	0.21(2)
132.7	0.81(2)	—	0.83(7)	0.90(8)	0.83(7)
162.7	25.3(3)	26.4(3)	28.4(9)	28.0(8)	25.5(5)
304.9	17.54(15)	17.67(18)	17.30(70)	17.8(5)	17.63(37)
423.7	12.65(12)	12.73(14)	12.80(60)	12.80(30)	12.79(62)
437.6	7.91(8)	7.82(9)	7.80(40)	7.80(25)	7.91(16)
467.7	0.029(3) 0.028(5)[a] 0.030(4)[b]	—	0.60(5)	<0.01	0.60(5)
537.3	100(1)	100(1)	100	100(1)	100
551.2	0.028(4)	—	—	0.027(9)	—

[a]Values measured using 96.0 cm^3 HPGe detector.
[b]Values measured using 57.1 cm^3 HPGe detector.

TABLE 3. Relative intensities of X rays and γ rays in a ^{140}La decay

Energy (keV)	Relative intensities					
	This work	Ref. 7	Ref. 9	Ref. 8	Ref. 10	Ref. 19
34.56 $\overline{K}_\alpha$	1.72(4)	2.35(70)	—	—	1.77(6)	1.67(6)
39.23 K_{β_1}	0.31(1)	0.36(8)	—	—	0.36(2)	0.31(2)
40.44 K_{β_2}	0.085(4)		—	—	0.089(4)	0.082(5)
64.14 γ	0.015(2)	0.01	—	—	0.011(4)	0.011
68.91	0.079(2)	0.064(16)	0.070(16)	—	0.080(6)	0.070(16)
109.4	0.230(4)	0.210(15)	0.170(10)	—	0.220(10)	0.209(15)
131.1	0.49(1)	0.50(3)	0.44(1)	—	0.48(1)	0.46(1)
173.5	0.133(4)	0.130(20)	0.120(10)	—	0.120(10)	0.135(21)
241.9	0.434(8)	0.410(30)	0.450(10)	—	0.460(30)	0.493(10)
266.5	0.488(8)	0.490(30)	0.520(10)	—	0.510(30)	0.474(26)
306.9	0.026(7)	0.035(17)	0.022(6)	—	0.020(5)	0.022(7)
328.7	21.1(3)	19.4(6)	21.5(4)	21.5(2)	21.7(4)	21.7(4)
397.5	0.077(5)	0.110(35)	0.078(3)	—	0.070(5)	0.105(31)
432.5	3.04(3)	2.85(15)	3.05(3)	3.08(3)	2.97(15)	3.13(6)
438.5	0.041(10)					
	0.046(18)[a]	0.021(10)	0.006(3)	—	<0.0014	0.021(11)
	0.038(12)[b]					
445.5	0.003(1)	≈0.025	0.005(3)	—	0.004(1)	0.025
487.0	47.7(6)	45.0(15)	46.6(9)	47.7(5)	46.4(9)	48.1(10)
618.1	0.039(4)	≈0.045	0.049(6)	—	0.014(3)	0.063(31)
751.6	4.54(4)	4.40(20)	4.45(5)	4.65(5)	4.36(16)	4.51(31)
815.8	24.4(2)	23.5(7)	24.0(4)	24.8(2)	23.5(7)	24.7(5)
867.7	5.77(7)	5.60(30)	5.69(6)	5.90(6)	5.56(19)	5.86(12)
919.5	2.79(3)	2.64(16)	2.83(4)	2.91(4)	2.80(9)	2.81(5)
925.2	7.23(7)	7.10(30)	7.26(8)	7.42(8)	7.10(21)	7.37(4)
950.9	0.544(7)	0.550(30)	0.553(7)	—	0.56(3)	0.567(20)
992.9	0.014(5)	—	—	—	0.009(3)	—
1045.0	0.026(15)	—	0.024(4)	—	0.016(4)	—
1097.2	0.024(5)	—	0.024(5)	—	0.022(5)	—
1303.5	0.044(7)	—	0.046(6)	—	0.050(7)	—
1405.2	0.062(7)	—	0.066(9)	—	0.068(8)	—
1596.2	100.0(15)	100.0	100.0	100(1)	100	100
1877.3	0.043(4)	—	—	—	0.042(6)	—
1924.6	0.014(2)					
	0.014(2)[a]	—	0.014(3)	—	0.006(2)	—
	0.013(3)[b]					
2083.2	0.031(2)					
	0.033(4)[a]	—	0.045(3)	—	0.007(2)	0.012(7)
	0.029(3)[b]					
2347.9	0.89(3)	0.90(6)	0.89(2)	0.89(2)	0.89(3)	0.89(2)
2464.1	0.012(2)	0.002(1)	0.012(1)	—	0.008(1)	0.018(6)
2521.4	3.63(4)	3.60(18)	3.58(15)	3.62(7)	3.61(9)	3.58(5)
2547.3	0.106(3)	0.110(7)	0.105(2)	0.109(3)	0.109(5)	0.109(3)
2899.6	0.070(2)	0.065(6)	0.070(1)	0.069(1)	0.069(3)	0.069(2)
3118.5	0.026(1)	0.027(4)	0.027(1)	0.027(1)	0.028(2)	0.027(1)
3320.4	0.0040(3)	0.0047(15)	0.0040(3)	—	0.0045(4)	0.0047(15)

[a]Values measured using 96.0 cm^3 HPGe detector.
[b]Values measured using 57.1 cm^3 HPGe detector.

different detectors in the coinciding regions, were found to be consistent with each other and the weighted averages of these results are given in Tables 2 and 3. The overall uncertainty in the intensity measurements includes the uncertainties due to statistics and peak-area evaluation (less than 0.5% for intense peaks), the efficiency-calibration error (1.5–2.5% over the energy region 4–100 keV and 0.5–2.0% over the energy region 100–4000 keV) and other systematic errors (0.3%). All of these uncertainties are added in quadrature to give the final uncertainties in the intensity results. The results of the earlier measurements are also included in the Tables 2 and 3 for comparison.

2.1.1. ^{140}Ba decay

The present measured γ-ray-intensity values in the ^{140}Ba decay agree well with the values reported by Gehrke *et al.* (8) and Adam *et al.* (10) (Table 2). The 551.2 keV γ ray reported only by Adam *et al.* (10) was seen clearly in our measurements and the intensity values are found to be consistent. The intensity

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of the 467.7 keV γ ray is found to differ remarkably from the earlier reported value (4). This may be due to a large summing of cascading γ rays falling under this peak (Table 1). The intensity values of K X rays and low-energy γ rays measured by us are found to be significantly lower than the earlier measured values of Kalinnikov *et al.* (4). However, the present intensity values of K X rays are more precise and show good agreement with the values measured by Adam *et al.* (10) as well as those calculated by using the physical parameters from internal-conversion and sequential atomic processes (19). The intensities of the different components of L X rays in the ^{140}Ba decay were measured by us for the first time and are found to be in good agreement with the evaluated results (19).

2.1.2. ^{140}La decay

In the case of the ^{140}La decay, the γ-ray intensities were measured separately from both the ^{140}La decay and from the successive decay of ^{140}Ba. The intensity values obtained from both of these measurements were found to be consistent and their weighted average is shown in Table 3. The present measured intensity values for most of the γ rays are in general agreement with the earlier measured values (5–10). The intensities of 438.5, 1924.6, and 2083.2 keV γ rays are found to differ significantly from the earlier reported values for these transitions (Table 3). The existence of 992.9 and 1877.2 keV γ rays in the ^{140}La decay, reported only by Adam *et al.* (10), was confirmed in the present measurements, and results are found to be in good agreement (Table 3). The present K X-ray intensities are about 20% lower than those measured by Kalinnikov *et al.* (7), but are in good agreement with those of Adam *et al.* (10) and the evaluated results (19). It is worth mentioning that the intensities of the L X rays in the ^{140}La decay are very small and could not be measured in our measurements.

In conclusion, we believe that the present results are more reliable and determined with better precision when compared with previous results because of the very good statistics collected under the weak peaks and the use of four well-calibrated semiconductor detectors.

3. Internal conversion-electron measurements

The mini-orange electron spectrometer used for the conversion-electron measurements comprises (*i*) a windowless Si(Li) detector (surface area = 200 mm^2, depletion depth = 3 mm, FWHM = 1.7 keV for 624.5 keV conversion electrons from ^{137}Cs decay); (*ii*) a mini-orange filter composed of five thin square (side = 1.5 cm) $SmCo_5$ permanent magnets fixed in an orange array in an aluminium circular frame (diameter = 6.0 cm). The filter has a lead absorber (diameter = 0.8 cm, thickness = 1.8 cm) at the centre to avoid direct exposure of the Si(Li) detector to the photons emitted from the source. A clean vacuum of 3×10^{-7} Torr (1 Torr = 133.3 Pa) was maintained in the spectrometer using a triode vacion pump. The detector was coupled to an 8K-channel analyser (ND 66B) through a spectroscopy amplifier (ORTEC 572, shaping time = 2 μs).

Thin and uncovered sources of ^{140}Ba and ^{140}La were prepared by drying the source solution on aluminized Mylar backing supported on a brass ring (diameter = 1.0 cm). The count rate for the different sources was kept between 500–1000 counts s^{-1} for all the measurements. The efficiency calibration curve for this spectrometer at source-to-filter and filter-to-detector distances = 2.8 cm (optimized for maximum efficiency in the energy range 200–1800 keV) was generated as described in our earlier paper (24). An overall uncertainty due to efficiency calibration was estimated to be 3.5% in the entire energy region 200–1800 keV.

Ten spectra were taken with each ^{140}Ba and ^{140}La source for time durations ranging from 30 000 to 300 000 s. A typical conversion-electron spectrum obtained with the ^{140}Ba source is shown in Fig. 2. The electron intensities were obtained by measuring the area under the conversion peaks and dividing it by the corresponding efficiency value at that energy.

The conversion-electron intensities for various transitions in ^{140}La and ^{140}Ce are presented in Tables 4 and 5, respectively. In the case of the ^{140}La decay, the conversion-electron intensities were measured from both ^{140}La and ^{140}Ba → ^{140}La → ^{140}Ce decays, and the results obtained were consistent with each other. A weighted average of two intensity results is shown in Table 5. The uncertainties shown in intensity values correspond to one standard deviation (1σ) and include uncertainties due to statistics and peak-area evaluation (less than 1.0% for intense peaks), efficiency-calibration error (3.5%), and other systematic errors (1.0%). All these uncertainties are added in quadrature to calculate the final uncertainties in the conversion-electron intensities.

3.1. ^{140}Ba decay

The conversion-electron intensities for various transitions from the ^{140}Ba decay are found to be in general agreement with the only available results of Adam *et al.* (10) as shown in Table 4. The present results show better precision when compared with the earlier measurements. The electron intensity for the (M + N.) conversion line of the 537.3 keV transition is reported for the first time.

The internal conversion coefficients for various transitions in ^{140}La were deduced using present-measured conversion-electron and γ-ray intensities. The γ-ray and conversion-electron intensities were normalized to yield the theoretically predicted α_K value (25) for the 537.3 keV ($1^- - 0^-$) pure $M1$ transition. The internal-conversion coefficients given by Adam *et al.* (10) are found to be, in general, higher than our values except for the 304.8 keV transition. The results quoted by Adam *et al.* (10) were normalized to yield the 162.3 keV ($2^- - 3^-$) transition to be pure $M1$, while present results were normalized to yield 537.3 keV ($1^- - 0^-$) transition to be pure $M1$. However, if we renormalize the results of Adam *et al.* (10) according to our values, the agreement with the present results is found to be reasonably good for all the transitions. The present measured conversion coefficients along with the theoretical values of Hager and Seltzer (25) are shown in Table 4. The multipolarities assigned to the different transitions in ^{140}La are shown in the last column of Table 4.

3.2. ^{140}La decay

In the case of ^{140}La decay, the present results for conversion-electron intensities agree well with the earlier measured results (5) (Table 5). The electron intensities for the (M + N) conversion of 328.8, 487.0, 1596.2, and 1903.5 keV transitions in ^{140}Ce are reported for the first time. The conversion coefficients for the various transitions in ^{140}Ce were calculated using the present conversion-electron and γ-ray intensity values and are presented in Table 5. The γ-ray and conversion-electron intensities were normalized to yield the theoretical α_K value (25) for the 1596.2 keV ($2^+ - 0^+$) pure $E2$ transition.

A comparison of measured values of K-conversion coefficients was made with those of the theoretical values of Hager

TABLE 4. Conversion-electron intensities, conversion coefficients, and multipolarities of various transitions in the decay of ^{140}Ba to ^{140}La

Energy of transitions (keV)	Shell	Internal conversion-electron intensities		Internal conversion coefficients (α)			Multipolarity
		This work[a]	Ref. 10[b]	Expt. this work ($\times 10^{-2}$)	Theoretical $M1$ ($\times 10^{-2}$)	Theoretical $E2$ ($\times 10^{-2}$)	
304.8	K	0.71(3)	0.62(3)	4.1(2)	4.4	3.6	$E2+M1$
423.7	K	0.23(2)	0.26(4)	1.8(2)	1.9	1.4	$M1$
437.6	K	0.12(1)	0.15(2)	1.5(1)	1.7	1.3	$M1$
537.3	K	1.04(4)	1.04(4)	1.04(4)	1.04	—	$M1$
	L	0.13(1)	0.15(2)	0.13(1)	0.14	—	
	$M+N$	0.057(8)	—	0.06(1)	—	—	

[a]Normalized to theoretical α_K value (25) for 537.3 keV ($0^- - 1^-$) pure $M1$ transition $\alpha_K(537) = 1.04 \times 10^{-2}$
[b]Renormalized to theoretical α_K value (25) for 537.3 keV pure $M1$ transition.

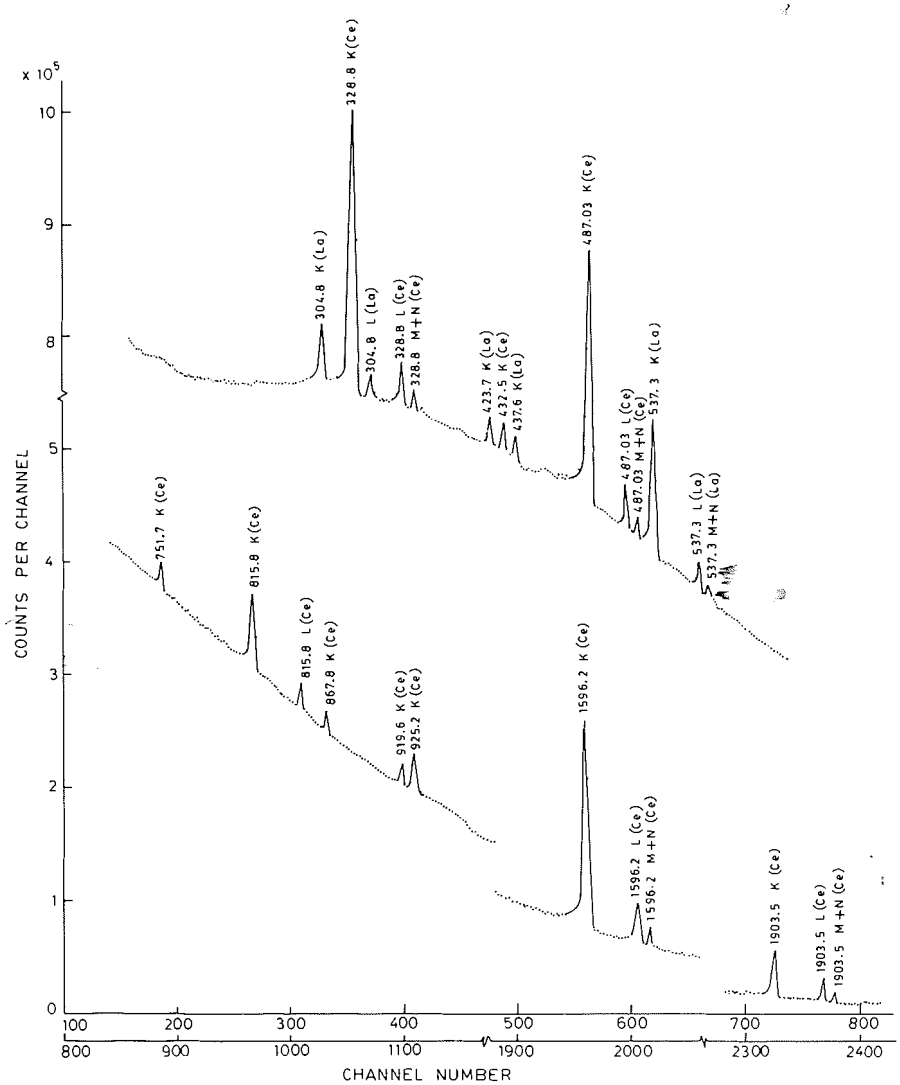


FIG. 2. Internal conversion-electron spectrum from ^{140}Ba decay taken with mini-orange electron spectrometer.

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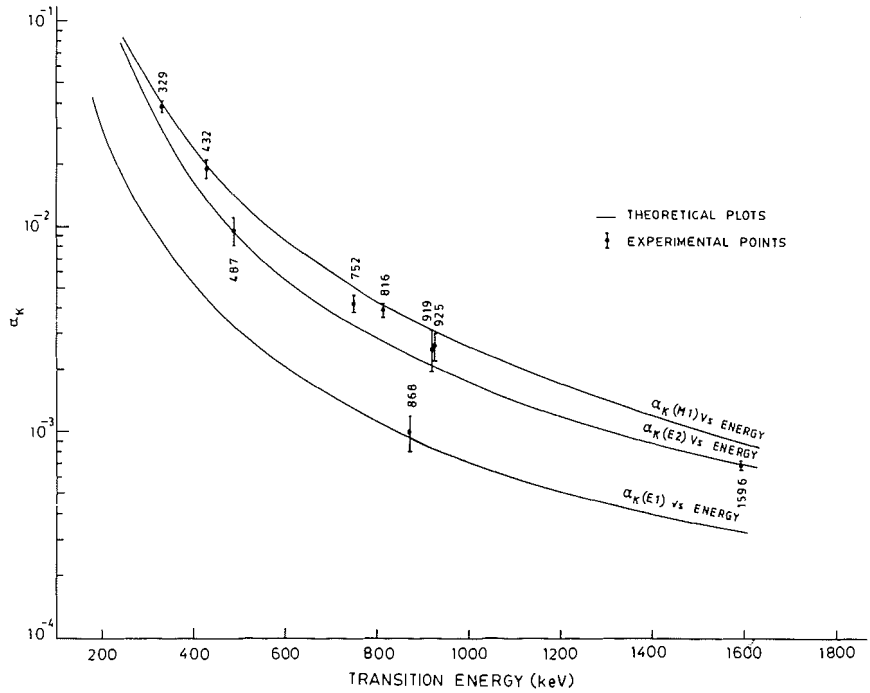


FIG. 3. Experimental K-conversion coefficients compared with the theoretical coefficients for $Z = 58$ given by Hager and Seltzer (25).

TABLE 5. Conversion-electron intensities, conversion coefficients, and multipolarities of transitions in decay of ^{140}La to ^{140}Ce

Energy of transitions (keV)	Shell	Internal conversion-electron intensities		Internal conversion coefficients (α)			Multipolarity
				Expt.	Theoretical		
		This work	Ref. 5	This work[a] ($\times 10^{-2}$)	$M1$	$E2$ ($\times 10^{-2}$)	
328.8	K	0.08(3)	0.76(1)	3.8(2)	3.92	2.98	$M1+E2$
	L	0.097(9)	0.102(5)	0.46(5)	0.52	0.55	
	$M+N$	0.021(3)	—	0.10(2)	0.11	0.12	
432.5	K	0.057(4)	0.058(3)	1.9(2)	1.94	1.34	$M1$
487.0	K	0.455(19)	0.440(23)	0.95(5)	$M3$; 12.9	0.92	$E2$
	L	0.062(5)	0.073(4)	0.13(1)	($M3$); 2.3	0.15	
	$M+N$	0.019(2)	—	0.040(4)	$M3$; 0.5	0.033	
751.7	K	0.019(2)	0.020(3)	0.42(4)	0.50	0.33	$M1+E2$
815.8	K	0.094(4)	0.095(6)	0.39(2)	0.41	0.27	$M1+E2$
	L	0.014(3)	0.014(1)	0.06(1)	0.05	0.04	
867.8	K	0.0058(9)	0.0061(15)	0.10(2)	$E1$; 0.11	$M2$; 0.92	$E1$
919.6	K	0.0070(15)	0.0061(15)	0.25(6)	0.31	0.21	$M1+E2$
925.2	K	0.019(2)	0.023(2)	0.26(3)	0.31	0.21	$M1+E2$
1596.2	K	0.069(3)	0.069(4)	0.069(3)	—	0.069	$E2$
	L	0.0071(10)	0.0084(10)	0.0071(11)	—	—	
	$M+N$	0.0012(3)	—	0.0012(3)	—	—	
1903.2	K	0.013(1)	0.015(2)	—	—	—	$E0$
	L	0.0013(2)	0.0018(3)	—	—	—	
	$M+N$	0.000 35(6)	—	—	—	—	

[a]Values normalized to α_K ($E2$) = 0.069 × 10^{-2} for the 1596.2 keV ($4^+ - 2^+$) pure $E2$ transition.

and Seltzer (25) for various possible multipolarities in Fig. 3. From this comparison multipolarities to various transitions in ^{140}Ce were assigned as shown in Table 5.

4. γ–γ directional correlation measurements in ^{140}Ce

For the directional correlation measurements, the sources were prepared in perspex holders that had a vertical cavity of 2 mm diameter and 3 mm depth. The count rates for these sources were kept at about 7000 counts s^{-1}. A 90 cm^3 HPGe – 96 cm^3 HPGe detector coincidence setup having a time resolution (2τ) = 7 ns for the 1173–1332 keV cascade following the ^{60}Co decay was used for these measurements. The energy resolution of both the detectors (1.7 keV at 1332 keV) was sufficient to completely separate the various γ rays from the ^{140}La

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TABLE 6. γ–γ directional correlation coefficients for various cascade in ^{140}Ce

Cascade (keV)	Spin sequence	Directional correlation coefficients A_{22}	A_{44}	Ref.
131–242	$4^+-5^+-6^+$	−0.28(13)	0.12(16)	This work
131–266	$4^+-5^+-4^+$	0.31(15)	−0.16(21)	This work
109–(329)–487	$2^+-(3^+-4^+)-2^+$	−0.31(2)	−0.14(2)	This work
266–487	$5^+-4^+-2^+$	−0.16(8)	0.14(11)	This work
		−0.10(3)	−0.05(4)	14
329–487	$3^+-4^+-2^+$	−0.102(9)	−0.012(11)	This work
		−0.105(9)	0.005(9)	14
		−0.092(12)	−0.037(23)	12
432–487	$(4^+)-4^+-2^+$	0.178(33)	0.073(45)	This work
	$(3^+)-4^+-2^+$	0.239(37)	−0.058(54)	14
		0.215(18)	−0.048(33)	12
329–(487)–1596	$3^+-(4^+-2^+)-0^+$	−0.081(9)	−0.029(11)	This work
		−0.096(6)	0.004(6)	14
		−0.124(21)	0.018(29)	9
		−0.110(5)	−0.041(8)	12
432–(487) − 1596	$(4^+)-(4^+-2^+)-0^+$	0.15(6)	0.06(8)	This work
		0.16(3)	0.09(5)	9
		0.26(1)	0.03(2)	13
		0.15(1)	−0.01(2)	11
	$(3^+)-(4^+-2^+)-0^+$	0.26(3)	−0.04(4)	14
487–1596	$4^+-2^+-0^+$	0.100(10)	0.021(15)	This work
		0.102(5)	0.018(7)	14
		0.097(8)	0.020(11)	9
		0.099(3)	0.002(8)	13
751–1596	$2^+-2^+-0^+$	−0.035(25)	0.072(33)	This work
		−0.022(18)	0.041(25)	14
		−0.024(33)	0.026(47)	9
		−0.030(36)	0.029(67)	13
815–1596	$3^+-2^+-0^+$	−0.097(8)	0.009(10)	This work
		−0.066(7)	0.001(9)	14
		−0.084(6)	−0.006(8)	9
		−0.097(6)	0.009(11)	12
868–1596	$3^--2^+-0^+$	−0.106(17)	−0.043(22)	This work
		−0.037(15)	−0.017(20)	14
		−0.084(16)	−0.048(22)	9
		−0.099(8)	0.007(10)	13
919–1596	$(4^+)-2^+-0^+$	0.145(52)	−0.114(69)	This work
		0.140(23)	0.009(34)	9
		0.130(80)	−0.02(12)	12
	$(3^+)-2^+-0^+$	0.077(35)	−0.035(53)	14
925–1596	$2^+-2^+-0^+$	0.386(21)	0.039(30)	This work
		0.322(23)	0.017(35)	14
		0.350(43)	−0.022(64)	9
		0.340(50)	0.010(70)	12
951–1596	$1^+-2^+-0^+$	−0.32(7)	0.009(9)	This work
		−0.27(7)	−0.07(11)	14
		−0.35(4)	0.21(5)	9
		−0.36(3)	0.008(51)	13

decay. The source to detector distance was kept at 15 cm for both the detectors. The γ rays of the 131, 487, and 1596 keV transitions observed by a movable 90 cm^3 HPGe detector were used for gating. The coincidence spectra were recorded at seven angles from 90 to 180° in intervals of 15°. The data were corrected for the miscentering of the source (<1.0%), the chance coincidences, and the Compton contributions. Also, the correlation coefficients were corrected for the finite resolution of the HPGe detectors by the method developed by Krane (26). The details of the experimental setup and method of analysis are described elsewhere (27, 28).

The results of directional correlation measurements for 15 cascades in ^{140}Ce along with the results reported by earlier workers are presented in Table 6. The directional correlation coefficients for 131–242, 131–266, and 109–(329)–487 keV cascades in ^{140}Ce were measured for the first time. The present measured values of the directional correlation coefficients for the 432–487, 816–1596, 868–1596, and 919–1596 keV cascades differ from the values reported by Michelkakis and Hamilton (14), but, are in agreement with the available results of Kaur *et al.* (9) and Saxena and Sharma (12).

The directional correlation coefficients from the present

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TABLE 7. Multipole mixing ratio (δ) for various transitions in ^{140}Ce

Cascade (keV)	Transition (keV)		Mixing ratio (δ) This work	Ref. 14	Ref. 9	Ref. 12	Multipole admixture (This work)
266–131	131		0.15 ± 0.18	$-0.13^{+0.02}_{-0.05}$	—	—	$M1 + (2.2^{+8}_{-2})\%E2$
242–131	242		$-0.60^{+0.35}_{-0.30}$	—	—	—	$M1 + (26.5^{+19.0}_{-17.7})\%E2$
266–487	266		-0.14 ± 0.12	20.2^{+43}_{-8}	—	—	$M1 + (1.9^{+5.4}_{-1.9})\%E2$
109–(329)–487	109		0.26 ± 0.02	—	—	—	$M1 + (6.5^{+1.0}_{-1.2})\%E2$
329–487	329		-0.04 ± 0.01	-0.049 ± 0.006	—	-0.04 ± 0.02	$M1 + (0.2 \pm 0.1)\%E2$
329–(487)–1596	329		-0.07 ± 0.01	—	$0.0^{+0.05}_{-0.0}$	-0.02 ± 0.01	$M1 + (0.5 \pm 0.14)\%E2$
432–487	432*	(*i*)	0.05 ± 0.01		—	$0.5^{+0.07}_{-0.03}$	$M1 + (0.25^{+0.17}_{-1.8})\%E2$
		(*ii*)	-0.41 ± 0.06	-0.54 ± 0.05		-0.8 ± 0.01	$M1 + (14.6^{+3.7}_{-3.0})\%E2$
432–(487)–1596	432*	(*i*)	$0.13^{+0.18}_{-0.15}$		-0.14 ± 0.06	$-0.5^{+0.1}_{-0.07}$	$M1 + (1.7^{+1.5}_{-5.6})\%E2$
		(*ii*)	-0.37 ± 0.08			-0.55 ± 0.10	$M1 + (12.0^{+6.0}_{-4.4})\%E2$
487–1596	487		-0.005 ± 0.02	—	$0.0^{+0.05}_{-0.0}$	0.0 ± 0.01	$E2 + (0.003^{+0.02}_{-0.05})\%M3$
751–1596	751		0.38 ± 0.04	0.36 ± 0.03	0.37 ± 0.06	$0.30^{+0.07}_{-0.02}$	$M1 + (13^{+2.3}_{-2.1})\%E2$
815–1596	815		-0.03 ± 0.01	0.01 ± 0.01	-0.0	-0.04 ± 0.01	$M1 + (0.11 \pm 0.01)\%E2$
868–1596	868		-0.044 ± 0.02	0.04 ± 0.02	-0.0 ± 0.05	0.25 ± 0.02	$E1 + (0.2^{+0.2}_{-0.15})\%M2$
919–1596	919*	(*i*)	0.07 ± 0.1	—	$0.07^{+0.05}_{-0.07}$	0.06 ± 0.09	$E2 + (0.6^{+2.7}_{-0.5})\%M3$
		(*ii*)	$1.9^{+0.5}_{-0.4}$	2.6 ± 0.4	—	$-0.08^{+0.06}_{-0.02}$	$M1 + (80 \pm 1)\%E2$
925–1596	925		-0.22 ± 0.04	-0.10 ± 0.04	$-0.16^{+0.07}_{-0.06}$	-0.15 ± 0.07	$M1 + (4.4^{+1.9}_{-1.4})\%E2$
951–1596	951		0.06 ± 0.07	0.01 ± 0.07	$0.06^{+0.03}_{-0.01}$	—	$M1 + (0.4^{+1.3}_{-0.4})\%E2$

*Values evaluated using the 2516 keV level spin (*i*) $I^{\pi} = 4^+$ and (*ii*) $I^{\pi} = 3^+$.

TABLE 8. Summary of multipole mixing ratios and *K*-conversion coefficients for 432 and 919 keV transitions in ^{140}Ce

Cascade (keV)	Spin sequence	Transition	Mixing ratio (δ)	Multipole admixture	α_K (calc.) Ref. 25	α_K (expt.)
432–487	$4^+ - 4^+ - 2^+$	432	$\delta_1 = 0.05 \pm 0.01$	$M1 + (0.25^{+0.17}_{-1.8})\%E2$	0.019	0.019(1)[a]
			$\delta_2 = -1.06 \pm 0.22$*	$M1 + (52^{+10}_{-12})\%E2$	0.016*	0.019(2)[b]
	$3^+ - 4^+ - 2^+$	432	$\delta_1 = -0.41 \pm 0.06$	$M1 + (14.6^{+3.7}_{-3.0})\%E2$	0.018	
			$\delta_2 = -4.84^{+0.97*}_{-1.42}$	$M1 + (96 \pm 2)\%E2$	0.013*	
432–(487)–1596	$4^+ - (4^+ - 2^+) - 0^+$	432	$\delta_1 = 0.13^{+0.15}_{-0.18}$	$M1 + (1.7^{+1.5}_{-5.6})\%E2$	0.019	
			$\delta_2 = -1.24^{+0.37*}_{-0.45}$	$M1 + (60^{+14}_{-17})\%E2$	0.015*	
	$3^+ - (4^+ - 2^+) - 0^+$	432	$\delta_1 = -6.0^{+1.9*}_{-4.9}$	$M1 + (97^{+2}_{-3})\%E2$	0.013*	
			$\delta_2 = -0.37 \pm 0.08$	$M1 + (12^{+6.0}_{-4.4})\%E2$	0.018	
919–1596	$4^+ - 2^+ - 0^+$	919	$\delta_1 = 0.07 \pm 0.10$	$E2 + (0.6^{+2.7}_{-0.5})\%M3$	0.0022	0.0025(6)[a]
			$\delta_2 = 1.38 \pm 0.30$*	$E2 + (66^{+8}_{-12})\%M3$	0.011*	0.0023(7)[b]
	$3^+ - 2^+ - 0^+$	919	$\delta_1 = 0.32^{+0.12*}_{-0.09}$	$M1 + (9.5^{+7}_{-4})\%E2$	0.0030*	
			$\delta_2 = 1.9^{+0.5}_{-0.4}$	$M1 + (80 \pm 1)\%E2$	0.0023	

*δ values rejected on the basis of ICC measurements.
[a]Values taken from present measurements.
[b]Values taken from Karlsson *et al.* (5).

measurements were further used to deduce the mixing ratios for various transitions in ^{140}Ce using the method described by Krane and Steffen (29), whose notations and sign conventions are followed throughout. For the present analysis the 487 ($4^+ - 2^+$) and 1596 keV ($2^+ - 0^+$) γ transitions in ^{140}Ce were assumed to be pure *E*2 in character (30). The deduced multipole mixing ratios for the various transitions in ^{140}Ce are given in Table 7, along with the results reported by earlier workers (9, 12, 14).

The mixing ratio of the 266 keV transition is found to differ from that of Michelkakis and Hamilton (14), but is in agreement with the value reported by Garcia Bermudez *et al.* (13) from polarization-correlation measurements. The present value of mixing ratio for the 816 keV transition is somewhat higher than the value measured by Michelkakis and Hamilton (14) whereas, it is comparable with the value reported by Saxena and Sharma (12). The important aspects of the results from present γ–γ directional correlation measurements are discussed in the following sections.

4.1. The 109–(329)–487 keV cascade

The cascade was attempted for the first time. The measured correlation coefficients after solid-angle corrections are shown

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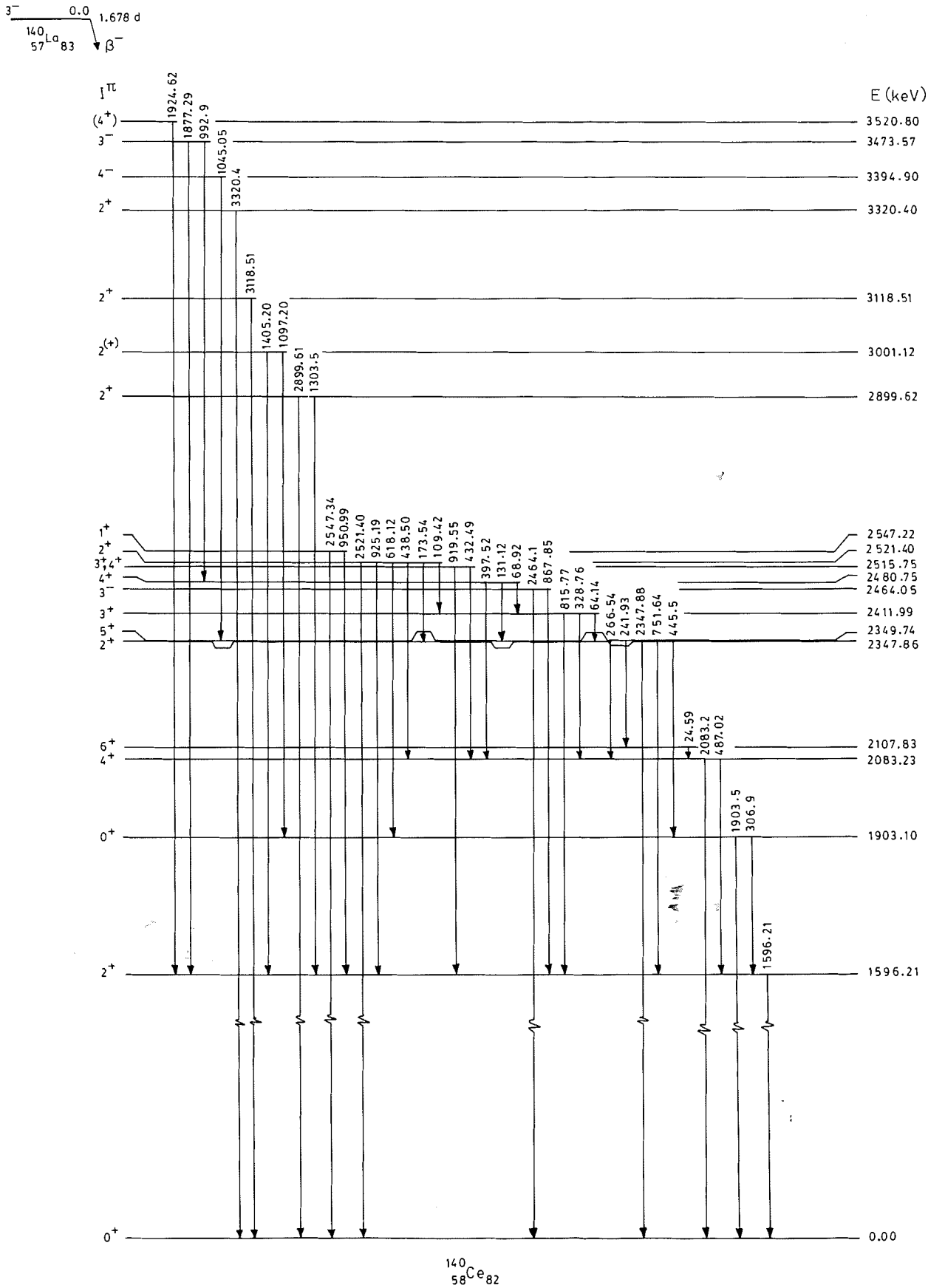


FIG. 4. Level scheme of ^{140}Ce from ^{140}La decay.

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in Table 6. This cascade follows a spin sequence $2^+-(3^+-4^+)-2^+$ having 3^+-4^+ spins for the unobserved 329 keV intermediate transition. Taking the 329 keV transition as $M1$ + 0.2% $E2$, from the present measurements, the U_k (329) coefficients were calculated using the tabulated values of Rose and Brink (30). Assuming the 487 keV transition to be pure $E2$ (31), the mixing ratio analysis for the 109 keV transition yields two values of the mixing ratio, δ(109),

$$\delta(109) = 0.26 \pm 0.02 \text{ and } \delta(109) = 1.85 \pm 0.10$$

Out of these two values, the value compatible with the measured internal-conversion coefficients (5) is $\delta(109) = 0.26 \pm 0.02$, which gives an admixture of $M1 + (6.5^{+1.0}_{-1.2})\%E2$ for the 109 keV transition.

4.2. Multipole admixture of the 266 keV transition

The measured A_{22} coefficient for the 266–487 keV cascade (spin sequence, $5^+ - 4^+ - 2^+$) in ^{140}Ce yields two possible values of mixing ratio for the 266 keV transition, i.e.,

$$\delta(266) = -0.14 \pm 0.12 \text{ and } \delta(266) = -17.5 \pm 5.5$$

The δ values and the measured K-conversion coefficients (5) indicate that the possible multipole components in the 266 keV transition are $M1$ and $E2$. However, it is not possible to select a unique δ value for the 266 keV transition on the basis of the measured α_K value (5) as shown below;

Mixing ratio	Multipole admixture	α_K calc. (ref. 25)	α_K expt. (ref. 5)
-0.14 ± 0.12	$M1 + (1.9^{+5.4}_{-1.9})\%\ E2$	0.068	0.063(10)
-17.5 ± 5.5	$E2$	0.057	—

On the other hand, the comparison of the measured K/L ratio = 6.5(10), see ref. 5, with the theoretical K/L values ($K/L(M1)$ = 7.5, $K/L(E2)$ = 4.5), see ref. 25, supports the multipolarity of the 266 keV transition as being of the predominantly $M1$ type. This is in contradiction to the 99%$E2$ component of this transition as reported by Michelkakis and Hamilton (14).

Also, if we use the above δ values of the 266 keV transition in the analysis of the 131–266 keV cascade, the $\delta(266) = -17.5 \pm 5.5$ gives the imaginary values of the mixing ratios for the 131 keV transition and is, therefore, ruled out. The value of $\delta(266) = -0.14 \pm 0.12$ gives two values of δ(131) viz.

$$\delta(131) = 0.15 \pm 0.18 \text{ and } \delta(131) = 3.1 \pm 1.2$$

Out of these two values, $\delta(131) = 0.15 \pm 0.18$, i.e., $M1 + (2.2^{+8.0}_{-2.0})\%E2$ admixture for the 131 keV is found to be compatible with the ICC measurements (5).

Thus, we can conclude from the measured K/L ratio for the 266 keV transition and from the directional correlation measurements of the 131–266 keV cascade that the multipolarity of the 266 keV transition is $M1 + (1.9^{+5.4}_{-1.9})\%E2$.

4.3. Spin of the 2516 keV level

The directional correlation measurements were made for three cascades, 432–487, 432–(487)–1596, and 919–1596 keV to assign a definite spin to the 2516 keV level. The mixing ratio analysis for the 432 and 919 keV transitions were done by taking the spin of the 2516 keV level to be 3^+ and 4^+; the values obtained are shown in Table 8. Also, in Table 8, K-conversion coefficients calculated using these mixing-ratio values and theoretical conversion coefficients from the Tables of ref. 25 by Hager and Seltzer are compared with the measured α_K values (5). Certain δ values were accepted on this basis. It is clear from this comparison that it is not possible to distinguish clearly between the 3^+ and 4^+ spin assignment to the 2516 keV level.

TABLE 9. Transition probabilities of various γ transitions in ^{140}Ce

Transition energy (keV)	Type	Transition probabilities: Single Particle	Transition probabilities: Experimental: This work	Transition probabilities: Experimental: Ref. 9
109	$M1$	3.76×10^{10}	1.49×10^{11}	—
109	$E2$	8.28×10^{5}	1.01×10^{10}	—
242	$M1$	4.11×10^{11}	2.18×10^{7}	—
242	$E2$	4.46×10^{7}	7.80×10^{6}	—
266	$M1$	5.46×10^{11}	3.37×10^{7}	—
266	$E2$	7.16×10^{7}	6.50×10^{5}	—
329	$M1$	1.03×10^{12}	5.60×10^{9}	5.7×10^{9}
329	$E2$	2.06×10^{8}	1.7×10^{7}	1.4×10^{7}
487	$E2$	1.47×10^{9}	4.96×10^{3}	5.0×10^{5}
816	$M1$	1.57×10^{13}	6.74×10^{9}	6.6×10^{9}
816	$E2$	1.94×10^{10}	7.35×10^{6}	—
868	$E1$	1.76×10^{15}	6.90×10^{12}	—
868	$M2$	1.10×10^{9}	1.33×10^{10}	—
925	$M1$	2.30×10^{13}	1.01×10^{13}	1.0×10^{10}
925	$E2$	3.65×10^{10}	4.87×10^{11}	2.6×10^{8}

The spin assignment to this level by the previous workers was done on the basis of weak arguments. Garcia Bermudez *et al.* (13) assigned a spin of 4^+ to this level from the $\gamma\gamma(\theta)$ measurements of the 432–(487)–1596 keV cascade. More recently, Michelkakis and Hamilton (14) assigned a spin of 3^+ from the $\gamma\gamma(\theta)$ measurements of the 432–487 and 432–(487)–1596 keV cascades. The value of A_{22} coefficients for the 432–(487)–1596 keV cascade measured by Garcia Bermudez *et al.* (13) and Michelkakis and Hamilton (14) are the same (Table 6), but these workers distinguished between two spin assignments on the basis of small A_{44} = 0.03(2) and −0.04(4) values measured by them. Further, no systematic trend was noticed in the A_{44} values measured by earlier workers (Table 6). The spin assignment of 4^+ by Kaur *et al.* (9) on the basis of comparison of ICC measurements is also found to be not very reliable.

5. Decay scheme and conclusions

The decay scheme of ^{140}La ($T_{1/2}$ = 1.68 d) based on the present and earlier measurements is shown in Fig. 4. The energies indicated in this figure are taken from ref. 31. The presence of the 1045.0 and 1924.6 keV transitions confirms the existence of the 3399.4 and 3520.8 keV levels, respectively. The level at 3000.7 keV is confirmed by the presence of γ rays of energy 1097.2 and 1404.9 keV. The presence of γ rays at 1877.3 and 992.9 keV confirm the existence of the level at 3473.6 keV in ^{140}Ce. The relative intensities of X rays and γ rays from the ^{140}Ba and ^{140}La decays were measured precisely and in some cases, these have been measured for the first time, using a set of well-calibrated, high-resolution Si(Li) and HPGe detectors. This work provides the conversion-electron intensities that have been determined with better precision.

The mixing ratios for the 109 and 242 keV transitions in ^{140}Ce have been determined for the first time from the directional correction measurements. These transitions are found to be almost $M1$ in character with a small admixture of $E2$. The

266 keV transition in ^{140}Ce is found to be dominantly $M1$, which is contrary to the accepted 99% $E2$ multipole admixture for this transition as given in ref. 31. The deduced multipole admixtures for the 329, 432, 487, 751, 816, 868, 919, 925, and 951 keV transitions, on the basis of γ–γ directional correlation measurements, are in good agreement with the values deduced from conversion-coefficient measurements as well as earlier reported values from reaction work (16).

The experimental transition probabilities for the 109, 242, 266, 329, 487, 816, 868, and 925 keV γ transitions in ^{140}Ce have been calculated using the present multipole mixing ratios and γ-ray-branching ratios. The values so obtained are compared with the earlier reported values (9) and with the single particles estimates in Table 9. The present values of transition probabilities for the 329 and 816 keV transitions show good agreement with the earlier reported values (9), whereas, significant differences are observed for the 487 and 925 keV transitions.

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